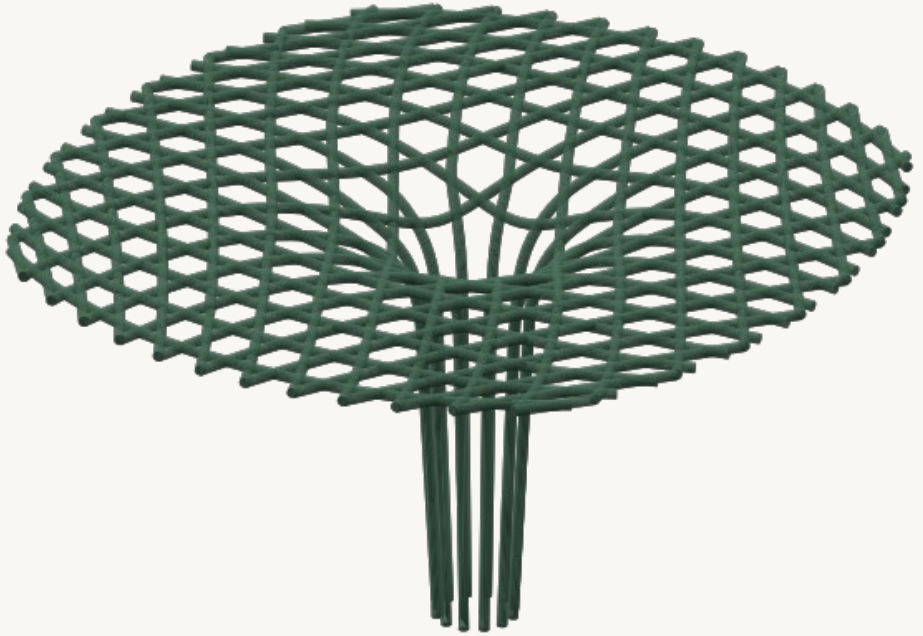
VineWeave Canopy

A tree-like climbing-plant support for pedestrian shade

Parametric design · Planting · Shade performance



TEAM & RESPONSIBILITIES

臧唯羽

Form & parametric modelling

林新淇

Materials & construction

虞寒哲

Shade performance & application

徐世程

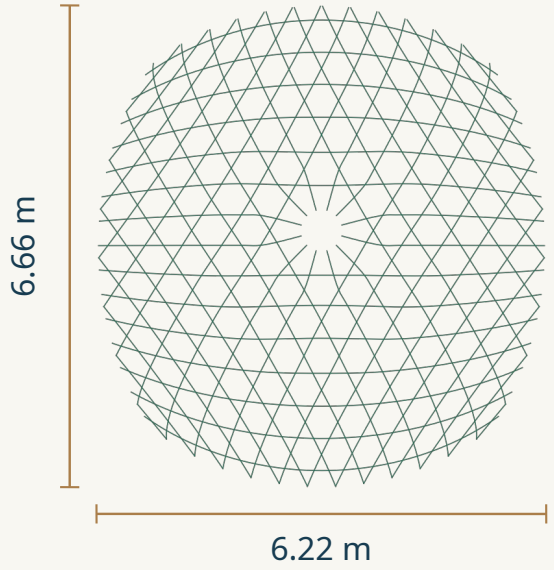
PPT production & integration

陈淼

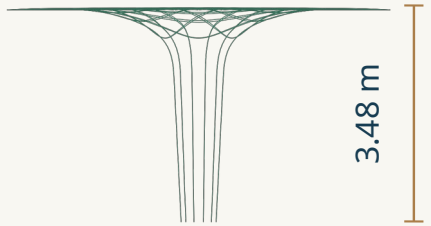
English presentation & Q&A

Define the final geometry—and how it is measured

Dimensioned plan/elevation views; a final parameter schedule, separate from sensitivity settings.



PLAN



ELEVATION

Curved support inclination is defined by its endpoint chord, not a local tangent.

FINAL PARAMETER	VALUE / DEFINITION
Overall height	≈3.48 m
Plan extents	≈6.22 × 6.66 m
Maximum horizontal span	≈6.91 m, exterior mesh
Support inclination*	45.9–50.7° to horizontal
Member exterior diameter	80 mm
Corner / expansion	0.85 / 1.08

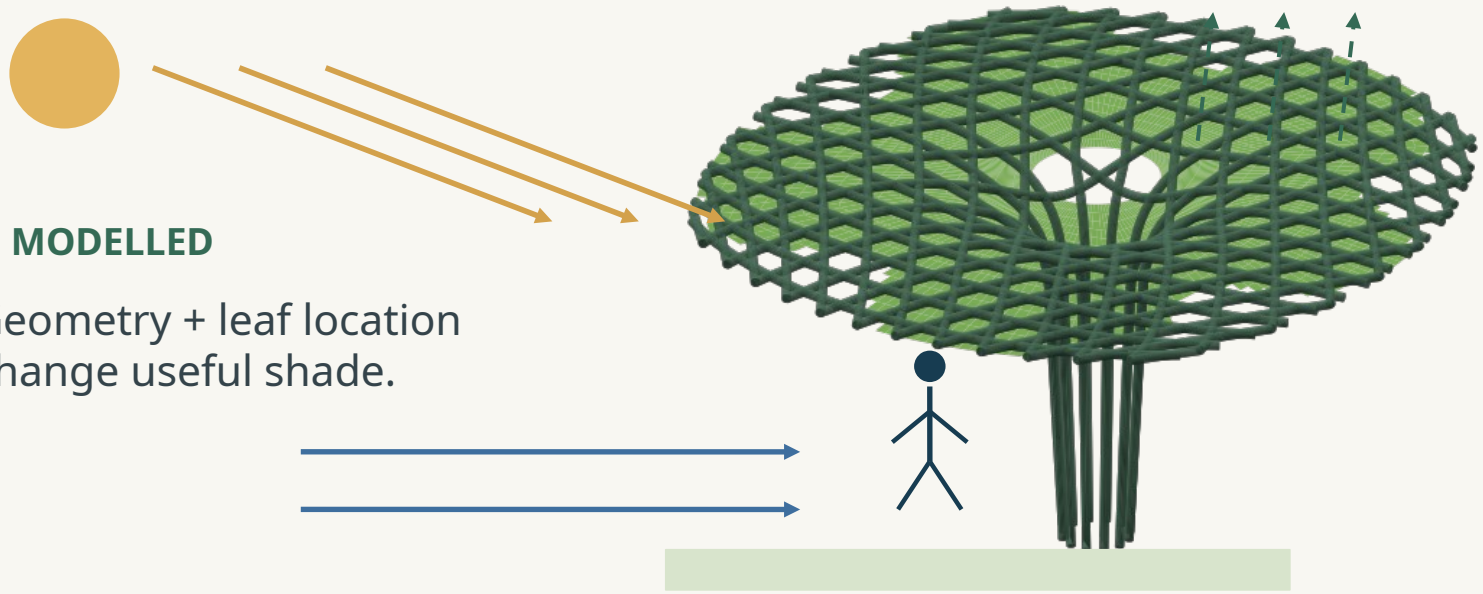
*Angle of each ground-reaching centre-line endpoint chord. Not a uniform member slope.

Model records P1 and supplemental geometric check. Centre-line views; exterior dimensions rounded. Drawing sheets accompany the PPT.

How the canopy can improve pedestrian comfort

Three physical pathways, with the strength of evidence shown next to each.

01 DIRECT SUN



MODELLED

Geometry + leaf location change useful shade.

03 PLANT WATER

Evapotranspiration may exchange latent heat; it needs healthy, watered plants.

02 AIRFLOW

Open sides retain possible air paths.
Dense foliage can also add resistance.

01 SHADE → LOWER EXPOSURE

Direct sun on people and ground

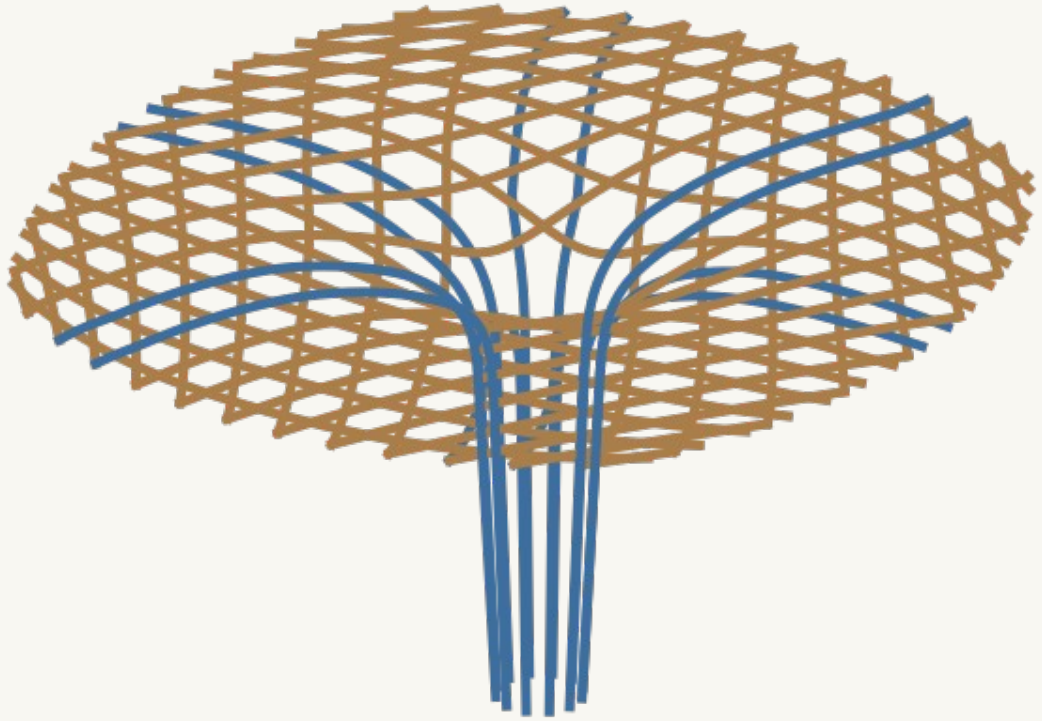
02 / 03: QUALITATIVE MECHANISMS

Shade % does not equal cooling °C.

Shade: project P2/P5. Greening mechanisms: [1,2]. Arrows are conceptual; no temperature or airflow field is simulated.

Assign materials to the roles they must perform

Steel at the ground-reaching supports; proposed engineered bamboo across the crown.



ZONE	PROPOSED SECTION	LENGTH
12 steel supports	Q355B tube Ø80 × 4 mm	69.38 m
47 bamboo crown members	Engineered bamboo solid Ø80 mm	219.28 m
TOTAL	59 members	288.66 m

Steel supports are proposed for force transfer and connection detailing. Crown bamboo requires stiffness, treatment and fabrication checks.

Project take-off and proposed member schedule (P1, P3). Sections and material mass remain preliminary specifications.

Read the material inputs before making the choice

Representative screening inputs: section resistance, conductivity, carbon factor and equal-length material cost.

MATERIAL / SECTION	BENDING* kN·m	λ* W/(m·K)	CARBON* $kgCO_2e/kg$	MATERIAL COST* CNY / 288.66 m
Q355B tube Ø80×4	6.14	50	2.15	18,394
Q355B tube Ø80×3	4.78	50	2.15	13,983
Engineered bamboo Ø80	1.01	0.22	0.40	10,158
Treated timber Ø80	0.85	0.14	1.10	6,530
Raw bamboo Ø80	0.40	0.18	0.25	4,330

DURABILITY	ENVIRONMENTAL ADAPTABILITY
Protect steel coatings; treat bamboo; detail inspection and replacement.	Check moisture, UV, wind exposure, local supply and plant suitability.

Source: material-comparison workbook (P3). *Preliminary inputs; product, processing and factor boundaries need verification.

Why steel supports and an engineered-bamboo crown?

Match structural roles first; weigh cost and environmental inputs together with practical constraints.

STEEL • 12 GROUND-REACHING MEMBERS

Prioritise stiffness and force transfer

The workbook assigns steel to supports under compression and wind bending. Its tabulated section stiffness EI exceeds the proposed bamboo section.

ENGINEERED BAMBOO • 47 CROWN MEMBERS

Use material benefits where feasible

The workbook adopts bamboo for the crown using cost and carbon inputs. It also requires restrained joints and verification of node rotational stiffness.

DURABILITY

Coat steel; protect and inspect bamboo.

COST

Include joints, installation and care.

CONDUCTIVITY

A material property, not a cooling result.

CARBON

Verify product factors and boundaries.

ADAPTABILITY

Check moisture, UV and local supply.

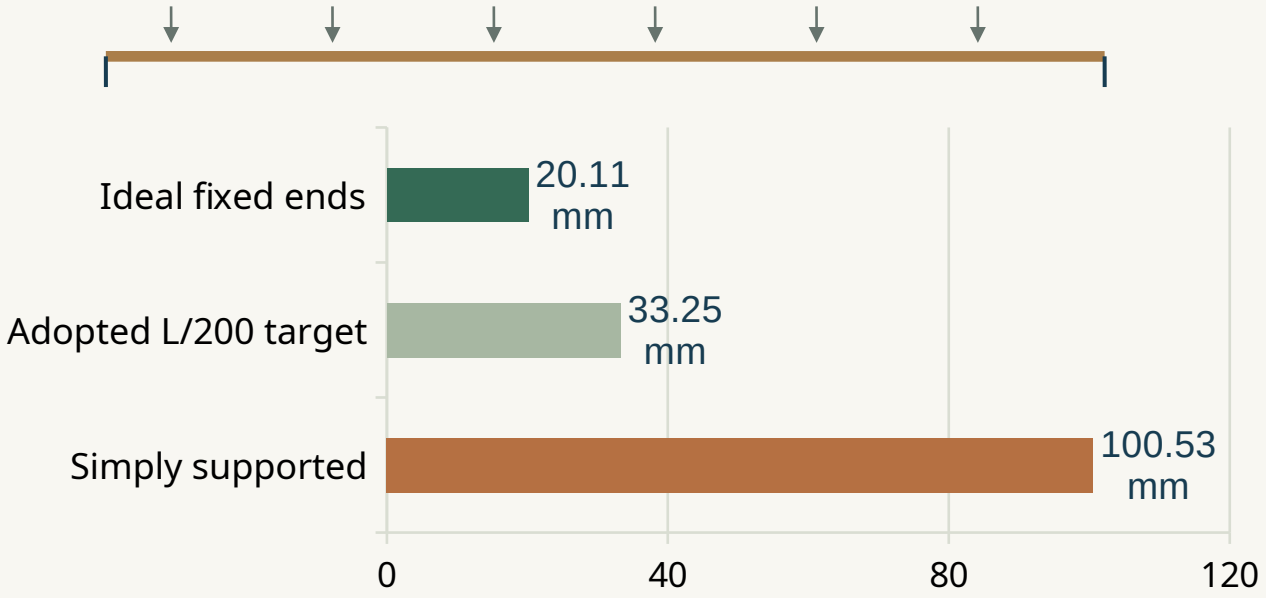
Source decision: bamboo crown + steel supports; node rotational stiffness still requires verification.

Design rationale P1/P3/P4. Screening values are preliminary; the hybrid strategy is not an optimised or certified structure.

The beam check sets a requirement for the joint

Same member and loading assumptions; different ideal end restraints produce different deflections.

SIMPLIFIED MEMBER: L = 6.65 m · SOLID Ø80 mm



Simple support: 3.02 × target | Ideal fixity: 0.60 × target

PROPOSED SUPPORT-HEAD NODE

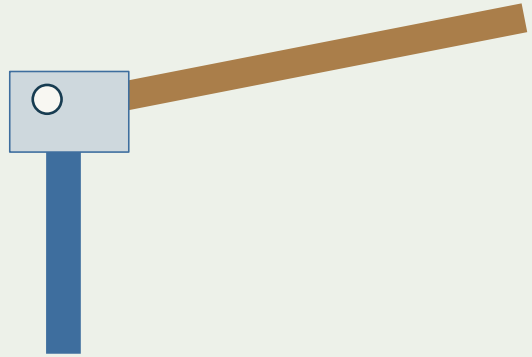


Plate + sleeve + adhesive + bolts

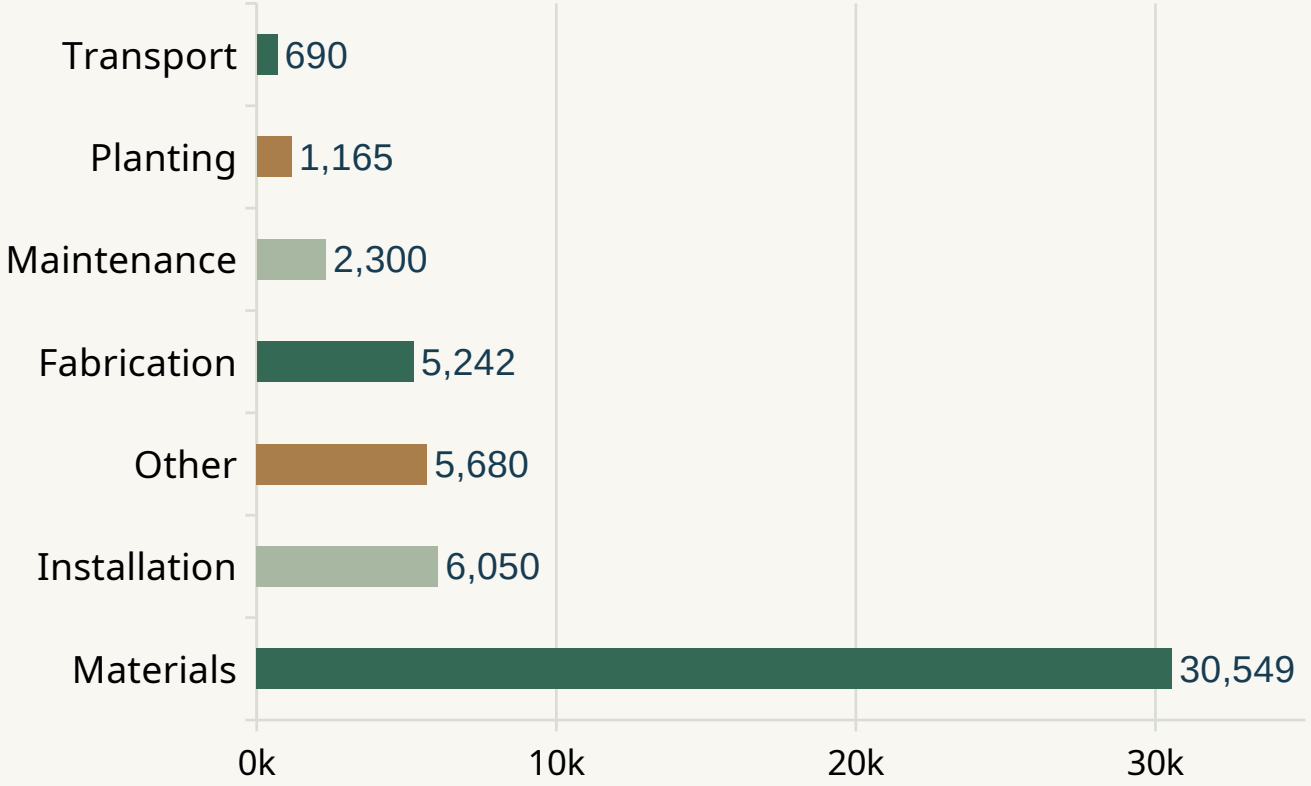
Concept: restraint is not yet demonstrated.

Next: verify joint rotation and whole-frame behaviour before claiming compliance.

P3 simplified beam check: self-weight + assumed 18 kg/m² wet foliage. E = 10 GPa and adopted L/200 need product/design confirmation.

The unit budget includes fabrication, installation and care

Preliminary total: CNY 51,676 per unit—not simply the price of the members.



MATERIALS = 59.1% OF TOTAL

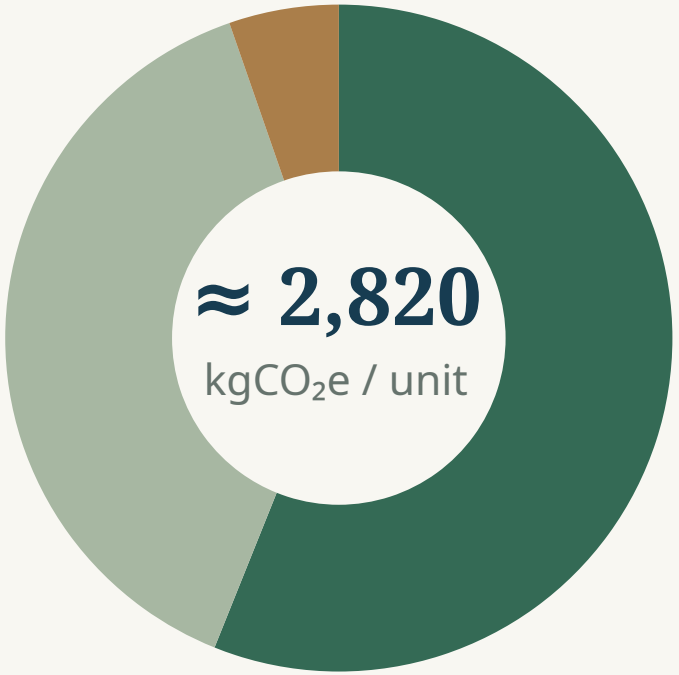
KEY COMPONENT	CNY
Bamboo crown	7,716
Steel supports	4,421
12 rigid-head nodes	12,000
Counterweight	3,402

These component rows are part of the full total. Nodes are a larger allowance than the primary steel members.

Source: Cost-Analysis.xlsx (P3). All rates remain reference estimates rather than confirmed quotations.

Account for the members, base and maintenance needs

Current carbon estimate: approximately 2,820 $kgCO_2e$ per unit.



CONTRIBUTOR	$kgCO_2e$
Primary members	1,581
Counterweight	1,089
Other components	150

Counterweight ≈ 39% of the current estimate

STEEL	BAMBOO	CONNECTIONS
Protect coating	Treat and inspect	Check fixings and grout

Workbook estimate (P3). Factor boundaries and node inclusion require checking; no biogenic-carbon credit is taken.

Keep the planting mix conditional on the site

Source proposal: 2 Boston ivy + 1 Rangoon creeper per unit; trumpet creeper is an alternative.

PRIMARY • 2 PLANTS

Boston ivy

Parthenocissus tricuspidata

Deciduous: summer leaf cover, winter leaf loss. Adhesive tendrils; train and check grip on the support.

COMPANION • 1 PLANT

Rangoon creeper

Combretum indicum

Source role: companion; twining climber. Confirm local suitability and attachment to the proposed rods.

ALTERNATIVE

Trumpet creeper

*Campsis grandiflora**

Vigorous woody climber; check roots, pruning demand and attachment. Confirm nursery species.

ROOTS + CARE

Source allowance: 1.2 m^3 substrate, drip irrigation, training and pruning.

Plant traits and quantities follow the workbook; local suitability and attachment require confirmation.

Two shade metrics answer different questions

30°N · assumed X=east/Y=north · true solar time 09–15 · same reference zone.

01	02	03	04
SUN DIRECTION	MODEL + LEAVES	SHADOW PROJECTION	TARGET INTERSECTION

01 IDEAL LEAF-SHADOW OVERLAP

Where could a continuous leaf envelope cast shade?

Leaf-envelope shadow on target ÷ target area. Used for the ideal daily / solar-season result.

02 EQUIVALENT DIRECT-BEAM BLOCKAGE

How much direct beam do the frame and assumed leaves block?

Accounts for frame/leaf overlap and assumed leaf transmittance $\tau = 0.15$; used for sensitivity tests.

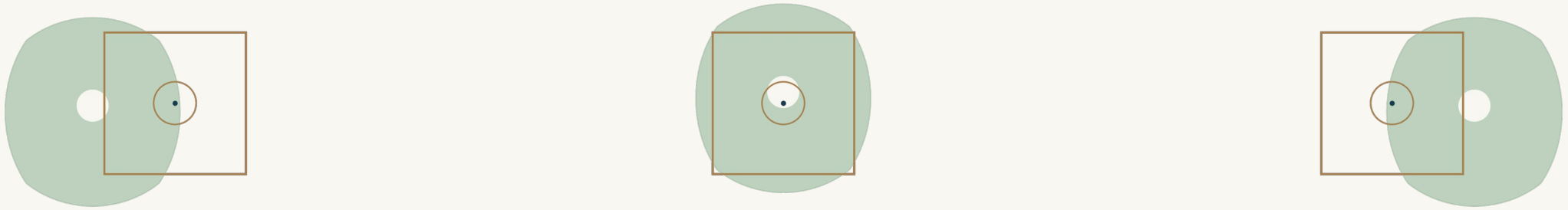
COMMON TARGET: 5 × 5 m MINUS r = 0.75 m → 23.23 m²

Compare results within the same metric and time window; do not add the two percentages.

Solar geometry: NOAA [3]. Project methods/assumptions P2/P5. Reference zone is not net usable floor area; no temperature change is calculated.

A noon peak does not ensure all-day shade

Metric 01: ideal leaf-shadow overlap. Read the moving footprint and the full 09–15 window.



09:00

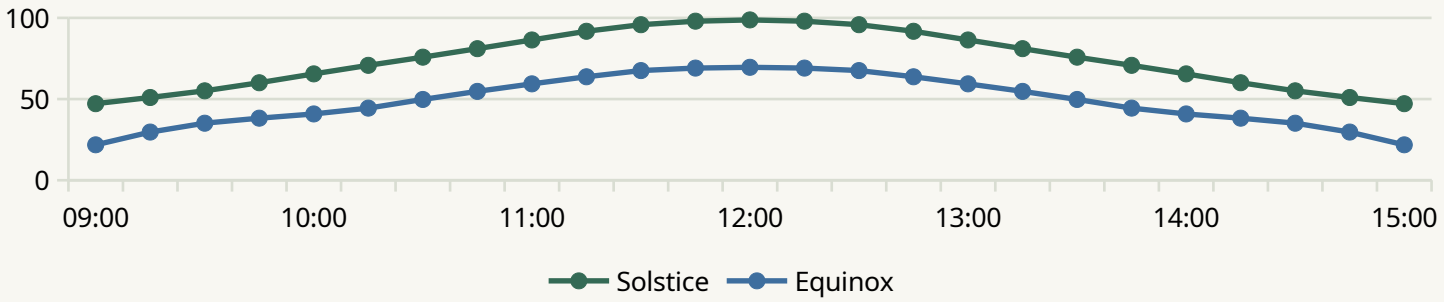
47.2%

12:00

98.7%

15:00

47.2%



09–15 TIME MEAN

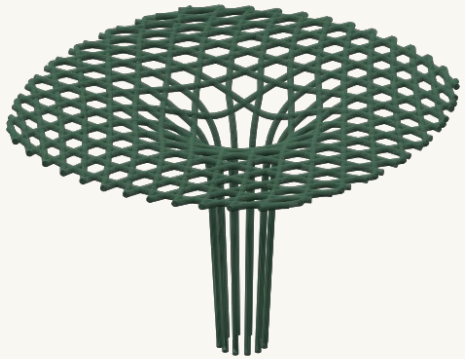
Solstice 75.3%
Equinox 49.8%

Design implication: check morning and afternoon seating exposure.

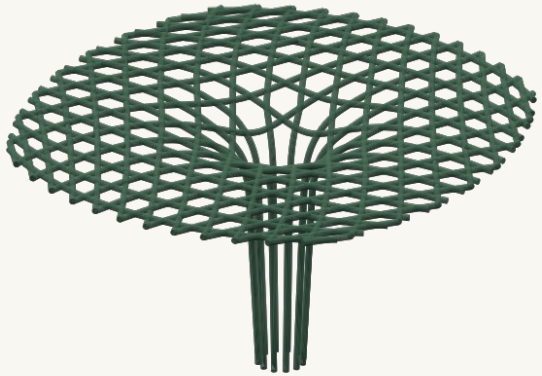
Project projection data (P2, P5). Solar-season comparison under the assumed foliage geometry.

Expanding the crown raises mean blockage: 52.3% → 66.1%

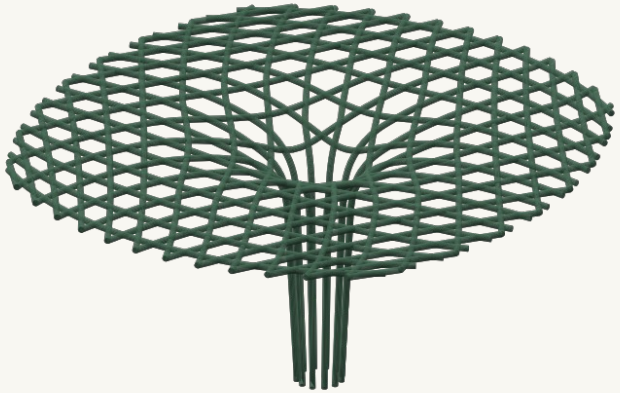
Metric 02: 09–15 mean blockage; one control at a time, 50% plan cover, solstice and τ = 0.15.



SMALLER CROWN



BASELINE



LARGER CROWN

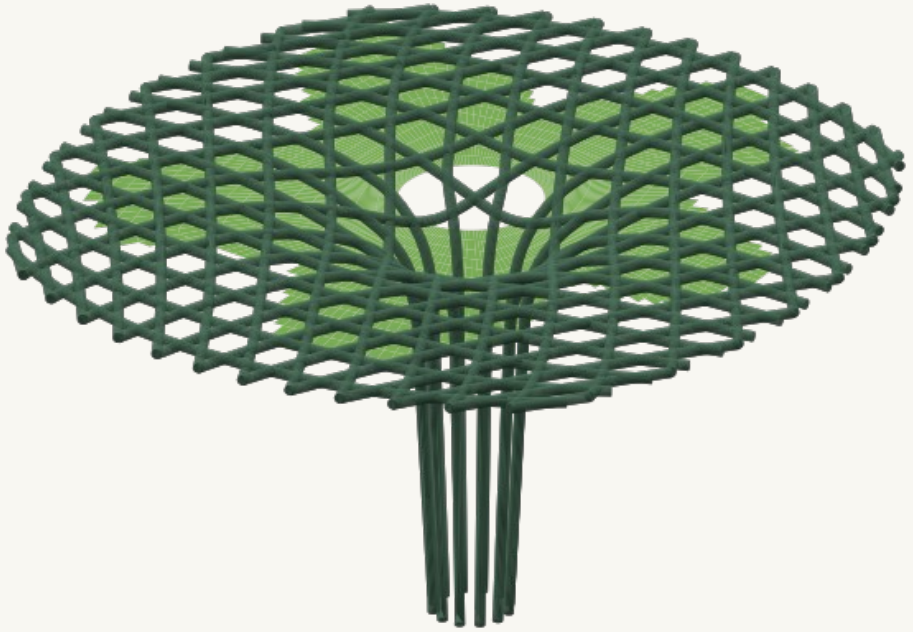
CONTROL	TESTED SETTINGS	MEAN BLOCKAGE: LOW / BASE / HIGH
Upper expansion	0.92 / 1.08 / 1.24	52.3% / 60.2% / 66.1%
Upper-height control	−0.4 / 0 / +0.4 m	62.6% / 60.2% / 57.8%
Corner factor	0.55 / 0.85 / 1.00	61.8% / 60.2% / 59.1%

Wider crown → more planted area at the same cover percentage; input quantities are not held equal.

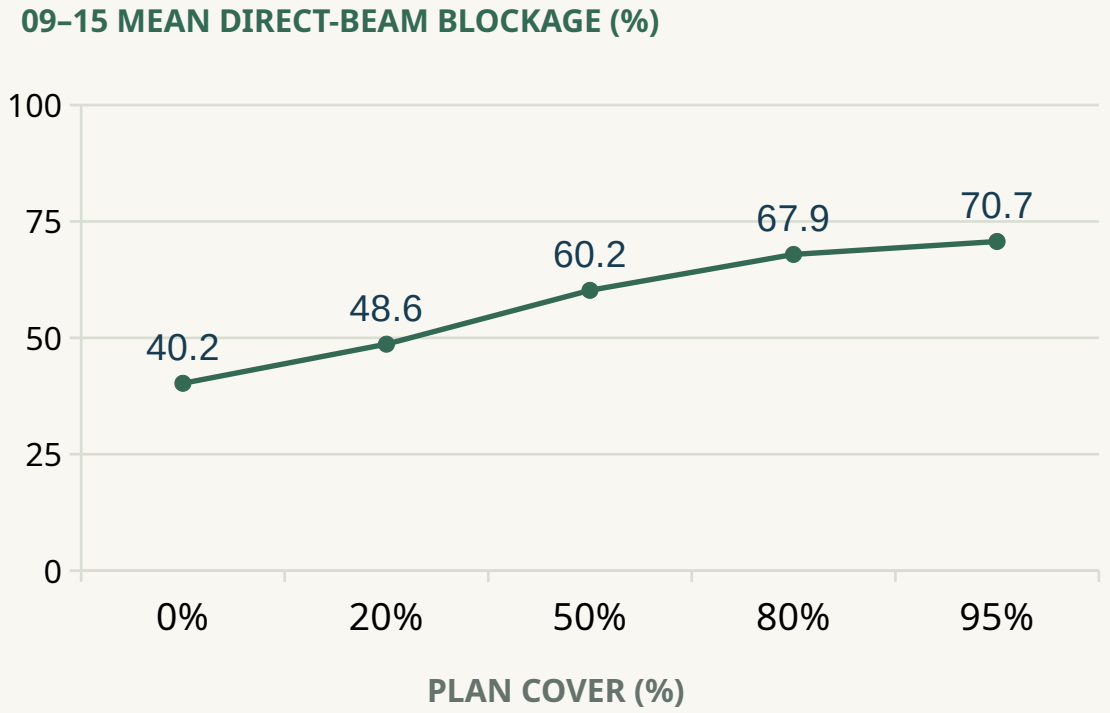
Project parameter study (P5). Controls are not uniform scaling; larger cover percentage does not imply equal plant/material input.

Coverage amount alone does not determine useful shade

Metric 02: fixed structure, five spatial-cover scenarios; 09–15 mean blockage under the same assumptions.



STATIC VIEW: 50% PLAN COVER • ANIMATION IN PPT



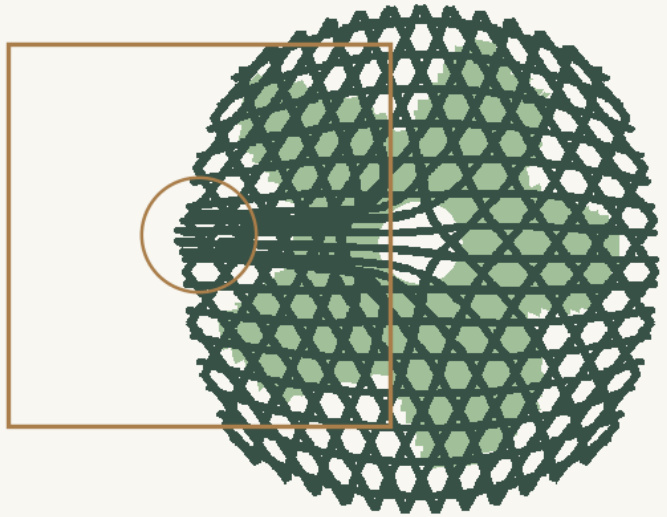
80 → 95% cover adds only 2.8 pp here.

Same 50% cover: noon blockage varies from 62.6% to 77.5%. Leaf location matters.

Project spatial-cover calculation (P5). Frame/leaf overlap deducted, τ = 0.15; equivalent direct-beam blockage, not cooling.

Put occupied-hour shade into the layout decision

A rest node must coordinate seating, through movement, roots and a substantial base.



STATIC VIEW: 15:00 • FIXED 50% PLAN COVER

01 SEATING LOCATION

Use the occupied-hour footprint, not just the crown outline or shadow centroid.

02 SPACE FOR ACCESS & ROOTS

Allow for circulation, counterweight, root zone, irrigation and maintenance.

03 VALIDATE THE OUTCOME

Confirm site and planting; compare radiation, temperature, humidity and wind.

Conceptual application P4; fixed-50%-cover projection P5. No real-site compliance or measured comfort result is claimed; method lead [4].

Coordinate prefabricated modules and site interfaces

Numbered components help assembly; long members, heavy base and grouted joints constrain it.

MODULE	WHAT IS PREPARED
Crown	47 formed/treatment-ready bamboo members
Supports	12 proposed galvanised steel members
Connections	Head nodes, sleeves, clamps and base flange
Foundation	Precast counterweight with anchorage
Planting	2 + 1 plants, substrate and drip provision

01 **Prefab, treat & number**

02 **Lift base and set out**

03 **Support, align & assemble**

04 **Grout, cure & recheck**

05 **Plant, train & restore**

14 person-days estimated ≠ 14 calendar days. The ≈9.07 t base controls lifting; grouted nodes limit lossless reversal.

Project member/assembly and cost schedule (P3, P4). Proposed detail and labour quantities, not a tested construction programme.

Connect geometry, construction and useful shade

Three design workstreams now have traceable outputs.

GEOMETRY

Rhino/GH model and parameters

Final form, drawings and material take-off

MATERIALS & BUILD-UP

Screening, sections, budget and modules

Proposed interfaces and implementation constraints

PERFORMANCE

Time-dependent shade and sensitivity

Geometry, cover and spatial-distribution effects

7 PARAMETER MODELS

5 COVER STATES

1,750 TIME-STEP RESULTS

Next: confirm products and joints, check plant suitability, and test the real site.

Synthesis of project records P1–P5. Product, joint and site-specific evidence are the next verification priorities.

References and project evidence

Method sources and project inputs are separate; drawings, brief and schedules accompany the deck.

[1] OUTDOOR SHADE & COMFORT

Watanabe et al. (2014). Evaluation of outdoor thermal comfort in sunlight, building shade, and pergola shade during summer in a humid subtropical region. Building and Environment, 82, 556–565.

[2] GREENING MECHANISMS

Hoelscher et al. (2016). Quantifying cooling effects of facade greening: Shading, transpiration and insulation. Energy and Buildings, 114, 283–290.

[3] SOLAR GEOMETRY

NOAA Global Monitoring Division. General Solar Position Calculations.
gml.noaa.gov/grad/solcalc/solareqns.PDF

[4] COMFORT METHOD

UTCI Calculator / COST Action 730.
www.utci.org/utci_calc.php
Method lead only; no project UTCI result is reported.

PROJECT RECORDS

P1 Model & take-off • P2 Ideal shade analysis • P3 Cost-Analysis.xlsx
P4 Proposed design & build-up • P5 Parameter and spatial-cover simulation

Project models, optical scenarios and estimates are identified separately from published literature.